

Java Programming – Comprehensive Syllabus

Course Overview

This course provides a comprehensive introduction to Java programming, covering core programming concepts, object-oriented programming (OOP), exception handling, collections, multithreading, database connectivity, GUI development, and Java application development. Students will build real-world applications using industry-standard tools and best coding practices.

Duration: 8–12 Weeks (50–70 Hours)

Prerequisites:

- Basic computer knowledge
 - No prior programming experience required
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Module 1: Introduction to Java (Week 1)

Topics

- Introduction to Java
- Features of Java
- Java Architecture (JVM, JRE, JDK)
- Installing JDK and IDE
- Structure of a Java Program
- Variables and Data Types
- Operators
- Input and Output
- Type Casting
- Comments and Naming Conventions

Practical Exercises

- Hello World Program
 - Basic Calculator
 - Unit Conversion Programs
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Module 2: Decision Making & Looping (Week 2)

Topics

- if, if-else, Nested if
- switch Statement
- for Loop
- while Loop
- do-while Loop
- Enhanced for Loop
- break and continue

Practical Exercises

- Grade Calculator
 - Leap Year Checker
 - Multiplication Table
 - Number Guessing Game
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Module 3: Methods, Arrays & Strings (Week 3)

Topics

- Methods
- Method Overloading
- Recursion
- Arrays
- Multidimensional Arrays
- String Class
- StringBuilder
- Command Line Arguments

Practical Exercises

- Matrix Operations
- String Manipulation
- Recursive Programs
- Menu-Driven Applications

Module 4: Object-Oriented Programming (Week 4)

Topics

- Classes and Objects
- Constructors
- this Keyword
- Static Members
- Encapsulation
- Access Modifiers
- Packages
- Inner Classes

Practical Exercises

- Student Management System
 - Employee Management System
 - Bank Account Simulation
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Module 5: Advanced OOP Concepts (Week 5)

Topics

- Inheritance
- Polymorphism
- Method Overriding
- Abstract Classes
- Interfaces
- Final Keyword
- Object Class

Practical Exercises

- Vehicle Management System
- Library Management System
- Shape Calculator

Module 6: Exception Handling, Collections & Generics (Week 6)

Topics

- Exception Handling (try, catch, finally, throw, throws)
- Custom Exceptions
- Collections Framework
- List
- Set
- Map
- Queue
- Generics

Practical Exercises

- Exception Handling Programs
 - Collection-Based Applications
 - Generic Class Implementation
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Module 7: Multithreading, File Handling & JDBC (Week 7)

Topics

- Threads
- Thread Lifecycle
- Synchronization
- File Handling
- Serialization
- JDBC
- Database Connectivity
- CRUD Operations

Practical Exercises

- Multithreading Programs
- File Management System

- Student Database Application
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Module 8: GUI Development & Introduction to Web Technologies (Week 8)

Topics

- Swing Basics
- Event Handling
- GUI Components
- Layout Managers
- Introduction to JavaFX
- Introduction to Servlets
- JSP Basics

Practical Exercises

- Calculator Application
 - Student Registration Form
 - Simple GUI-Based Project
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Module 9: Mini Project (Week 9)

Choose one:

- Library Management System
 - Banking Management System
 - Student Information System
 - Hospital Management System
 - Inventory Management System
 - Employee Payroll System
 - Hotel Reservation System
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Module 10–12: Capstone Project

Complete Java Application Development

- Requirement Analysis
 - Object-Oriented Design
 - Database Design
 - User Interface Development
 - CRUD Implementation
 - Exception Handling
 - Testing and Debugging
 - Documentation
 - Final Presentation
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Tools & Software

- Java JDK
 - IntelliJ IDEA
 - Eclipse IDE
 - Apache NetBeans
 - Visual Studio Code
 - MySQL
 - MySQL Workbench
 - Git & GitHub
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Core Concepts Covered

- Java Programming Fundamentals
 - Object-Oriented Programming (OOP)
 - Methods, Arrays, and Strings
 - Exception Handling
 - Collections Framework
 - Generics
 - Multithreading
 - File Handling
 - JDBC
 - Swing & JavaFX Basics
 - Introduction to Servlets and JSP
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Assessment Scheme

Component	Weightage
Assignments	20%
Lab Exercises	20%
Quizzes	10%
Mini Project	20%
Capstone Project	30%

Learning Outcomes

By the end of the course, learners will be able to:

- Understand Java programming fundamentals and object-oriented programming principles.
- Develop modular applications using classes, objects, inheritance, polymorphism, interfaces, and packages.
- Handle exceptions and perform file operations for robust applications.
- Use Java Collections Framework and Generics for efficient data management.
- Develop multithreaded applications and implement synchronization techniques.
- Connect Java applications to relational databases using JDBC and perform CRUD operations.
- Build desktop applications using Swing and understand the basics of JavaFX, Servlets, and JSP.
- Debug, test, and optimize Java applications using industry-standard development tools.
- Design and develop real-world Java applications following software engineering best practices.